



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SALEM DISTRICT LORRY OWNERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

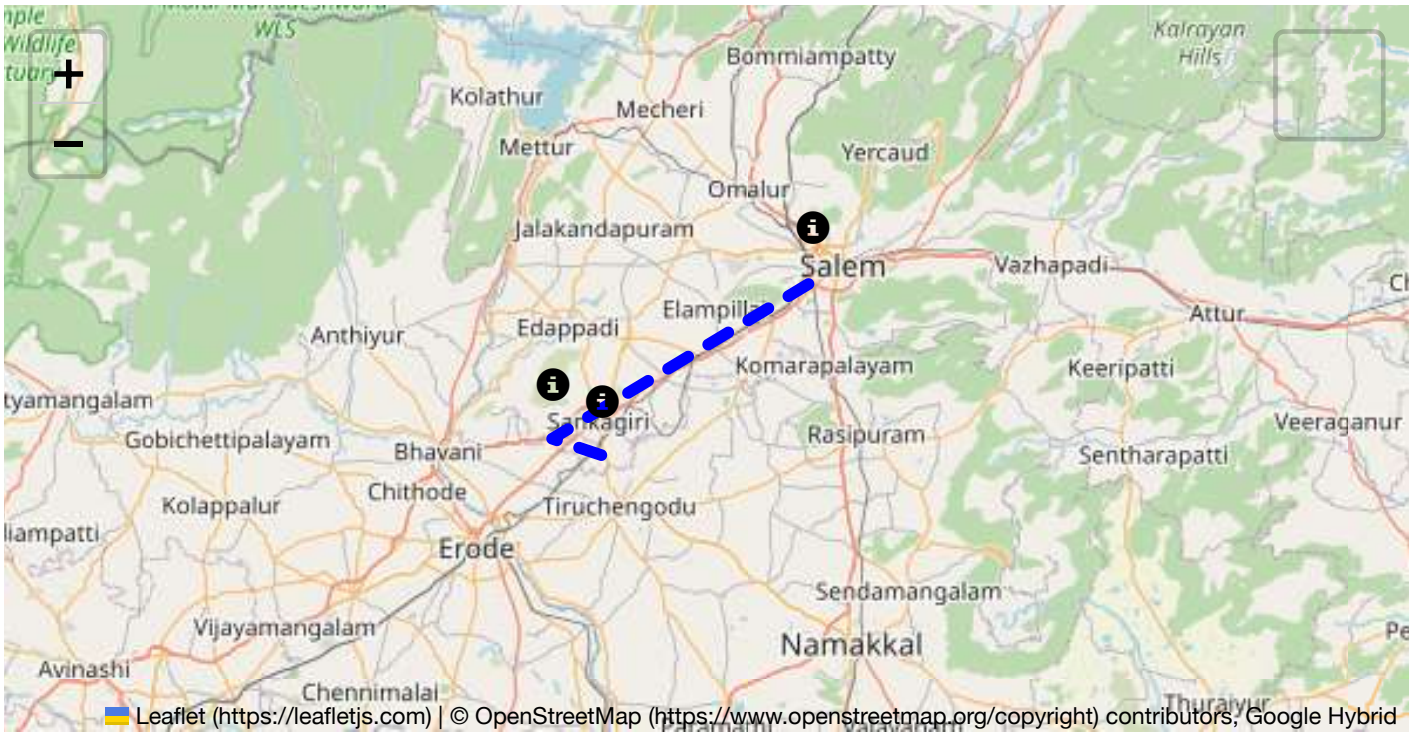
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

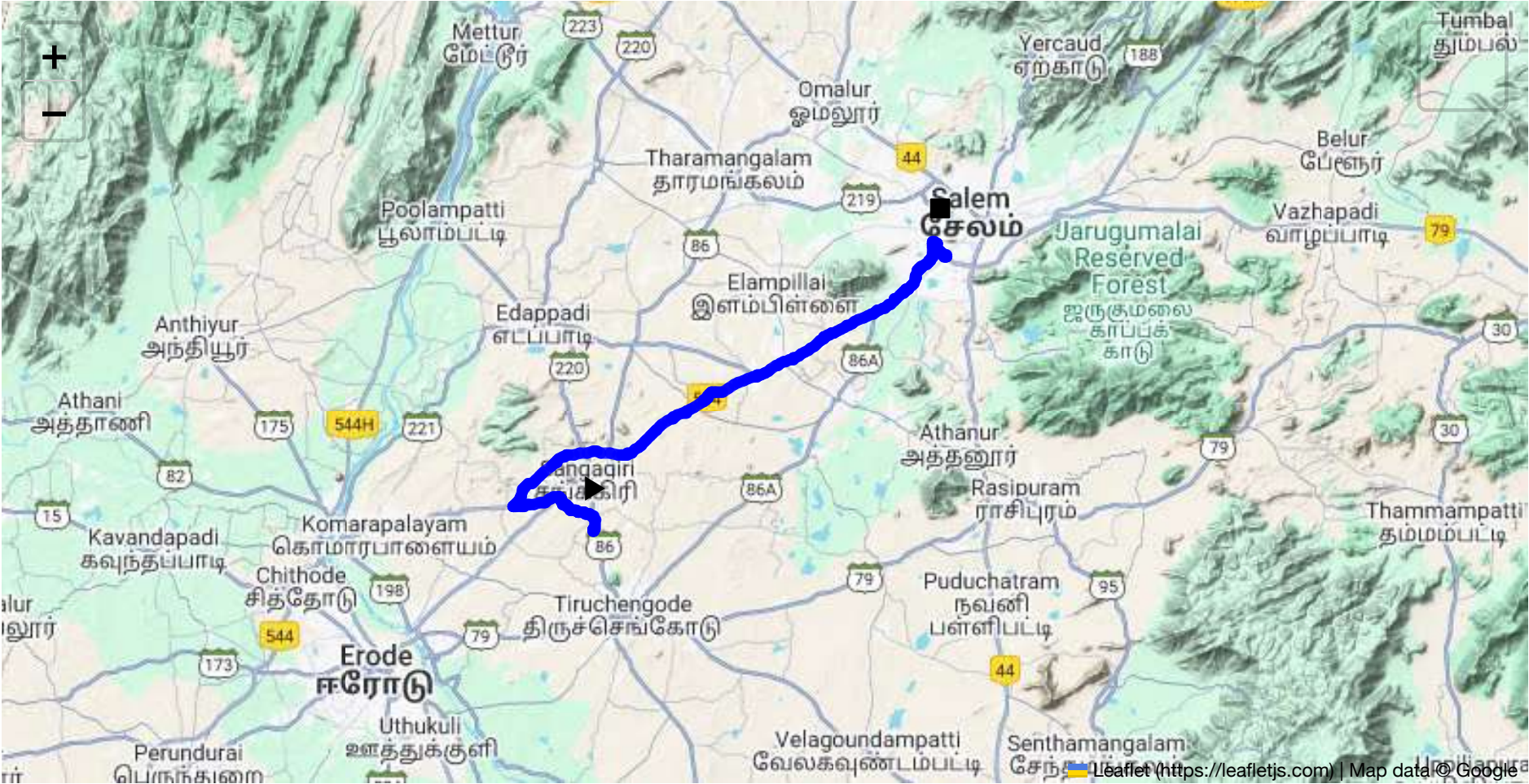


Route Summary:
Total Distance: 52.12 km
Estimated Duration: 1.0 hours

Adjusted Duration (Heavy Vehicle): 1.3 hours

Start: (11.4381, 77.8734)

End: (11.636291, 78.122116)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W (Sangagiri) to 205/8, Sandaipet, Periyakondalampatty, with an intermediate point at FR48+7XQ (Bhavani Main Road, Padaiveedu), covers approximately 52.12 km. It primarily involves major highways and arterial roads, making it suitable for heavy vehicles. This journey typically takes around 1 hour under normal conditions.

Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate with heavy monsoon rains from October to December, which can lead to waterlogging, reduced visibility, and slippery road conditions. Summer months often bring high temperatures and the potential for heat-related vehicle issues.

Traffic Patterns

- Peak Hours:** Expect increased traffic from 8:00 AM to 10:00 AM and 5:00 PM to 8:00 PM.
- Congestion-Prone Areas:** Bhavani Main Road and major junctions near Salem can experience slowdowns, particularly during peak hours.

Road Quality and Infrastructure

The highways and main roads are generally well-maintained, but be cautious of occasional potholes and narrow segments in rural areas. Construction or maintenance work may occur sporadically, requiring vigilance.

Alternative Routes for Emergencies

In case of major roadblocks, an alternative route could be via NH44 and SH86, although this may add distance and time. Check real-time traffic updates for the best detour.

Local Regulations on Hazardous Material Transport

Be aware that restrictions may apply to hazardous material transport, including time-of-day limitations and specific permitted routes. Adherence to Tamil Nadu's transport regulations is mandatory.

Historical Incidents

There have been few reported incidents involving heavy vehicles in recent years, mostly due to unauthorized parking or breakdowns. No significant hazardous materials incidents have been noted on this route.

Environmental Considerations

- **Sensitive Areas:** Be mindful of agricultural lands and small communities along the route, where noise or chemical spills could impact the local environment.
- **Wildlife Crossings:** Rare, but possible in rural sections. Proceed with caution.

Communication Coverage

Mobile network coverage is generally strong along the route, but expect potential dead zones in rural segments or where service providers differ.

Estimated Emergency Response Times

- **Urban Segments (Salem vicinity):** Approximately 15-20 minutes.
- **Rural Areas (Near Padaiveedu):** May extend up to 30-45 minutes due to distance and road conditions.

Overall Risk Assessment

- **Low to Moderate Risk:** The route is relatively straightforward for experienced drivers, with most hazards stemming from weather or peak traffic conditions.
- **Key Recommendations:** Monitor weather forecasts, check traffic updates frequently, and ensure vehicle readiness, especially in terms of braking and cooling systems.

Safe driving practices and adherence to regulations remain essential for transporting hazardous materials along this route. Regular communication with logistics and emergency services is recommended.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	Medium	11.45839, 77.85143	30 KM/Hr
7	Turn	Medium	11.46307, 77.84941	30 KM/Hr
8	Turn	Medium	11.46316, 77.84921	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
11	Turn	Medium	11.64156, 78.12162	30 KM/Hr
12	Turn	Medium	11.64116, 78.12168	30 KM/Hr
13	Turn	Medium	11.63270, 78.12757	30 KM/Hr
14	Turn	Medium	11.63256, 78.12749	30 KM/Hr
15	Turn	Medium	11.63247, 78.12734	30 KM/Hr
16	Turn	Medium	11.63252, 78.12712	30 KM/Hr
17	Turn	High	11.63640, 78.12244	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64156, 78.12162



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64116, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63270, 78.12757



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63256, 78.12749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63247, 78.12734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63252, 78.12712



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63640, 78.12244

